Yacoub Derradji

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Education

Western University Sept. 2024 – Apr. 2029 (Expected)

BESc, Mechatronic Systems Engineering, Dual in AI Systems Engineering *London, ON*

- Cumulative GPA: 3.8 (Dean's Honour List)
- Coursework: Dynamics, Mechanics of Materials, Manufacturing Processes, Fluid Mechanics, Heat Transfer, Computational Methods, Algorithms & Data Structures, Software System Design
- In progress: Introduction to Machine Learning (DS 3000, Fall 2026)

Experience

Undergraduate Researcher May 2026 – Present

Advanced Robotics & Mechatronics (ARM) Lab, Western University *London, ON*

- Redesigned 7 parts of a miniaturized planetary-gear magnetorheological (MR) clutch used as a per-joint actuator in a fully actuated multi-finger robotic hand, where one motor drives multiple clutches
- Traced a torsional shaft fracture and an unmanufacturable keyway to a shared root cause; resolved both by upsizing the shaft (1 to 2 mm) and implementing a D-profile shaft/bore coupling recommended by a labmate
- Relocated the electromagnetic coil radially outward while preserving clutch torque, justified from the magnetic circuit (magnetomotive force $N \cdot I$ is independent of coil radius) and by keeping the steel section that carries the flux
- Produced the 7-drawing SolidWorks manufacturing set under ISO 2768-m, catching a bearing-seat fit error during design review and re-specifying the seat to +0.02/0 mm
- Sourced all 7 parts across five manufacturing platforms, vetting vendors on gear-cutting capability and material compatibility; quotes pending approval for order

Planning and Controls Developer Oct. 2025 – Dec. 2025

Western Engineering Autopilot (WEAP) *London, ON*

- Made full-system ROS 2/Gazebo simulation startup 6x faster (3 min to 30 s) by implementing a structured launch architecture for the planning and control subsystems
- Integrated planning and control packages through coordinated topic publishing and deterministic startup sequencing, enabling repeatable simulation testing
- Extended a Python control console with live velocity control and safe shutdown for simulation testing

Electrical Team Member Sept. 2024 – Mar. 2025

Western Formula Racing (FSAE) *London, ON*

- Created the tractive system schematic and the tractive system connector pinouts for the team's electric race car

Leadership & Activities

Mechanical Lead, BotBrawl Combat Robot Team Aug. 2026 – Present

WE First Club, Western University *London, ON*

- Lead mechanical design and build of the club's combat robot for the 2026–27 BotBrawl season, organizing the mechanical sub-team

General Member Sept. 2025 – Aug. 2026

WE First Club, Western University *London, ON*

- Developed the neck-tilt mechanism for Baker Bot, the club's humanoid robot (see Projects); took part in the club's Ri3D build

Open House Volunteer Oct. 2025

Faculty of Engineering, Western University *London, ON*

- Spoke with prospective students and parents about the engineering program and student life

Projects

Humanoid Neck-Tilt Mechanism (Baker Bot) | *Onshape, Prototyping* May 2026 – Aug. 2026

- Developed a 2-DOF neck-tilt mechanism for Baker Bot, a torso-up humanoid (WE First), taking it from concept to working hardware: two angled 40 kg servos drive ball-jointed pushrods against a rear fixed link for pitch and limited roll
- Did the full Onshape design (off-the-shelf rod ends, custom 3D-printed mounts), prototyped and iterated it, implemented basic nodding, and integrated it with a teammate's rotation base before handing off to the software and torso teams

Off-Road RC Vehicle | *SolidWorks, Mechanism Design, Embedded Control* Dec. 2025 – Apr. 2026

- Designed the full vehicle in SolidWorks (chassis, modular FWD/4WD drivetrain, steering, suspension) and built it from custom 3D-printed parts under a $200 budget, with no kit
- Integrated Arduino control with motor drivers and servo steering, with PWM motor control, steering calibration, and a serial command interface
- First test drives exposed suspension and steering problems that kept it from driving; revised suspension arm lengths and steering knuckles to get it running

Tendon-Driven Robotic Gripper | *Prototyping, Grasping* Sept. 2025 – Present

- Built a 3-finger gripper prototype: adapted an open-source finger design for rubber-band and string tendons, and designed a 3-servo palm with one finger opposing two

Research Paper Summarizer | *Python, SQLite, Pydantic, LLM APIs* 2026

- Built a Python tool that summarizes research papers with an LLM, using a provider-agnostic backend (Ollama or Anthropic API), Pydantic data models, SQLite storage, and a web interface

CityLaunch (SE 2251 Course Project) | *Java, REST APIs, UML* Winter 2026

- Co-developed a Canadian city recommendation system for engineering graduates combining data from five APIs (Adzuna, GitHub, Statistics Canada WDS, OpenCorporates, Numbeo)
- Designed the full UML class, activity, and sequence diagrams for the system

TTC Flood Rerouting Simulation (ES 1050) | *Python, Graph Algorithms, GIS* Winter 2025

- First-year team design project: simulated how TTC service reroutes around flooded streets using GTFS schedule data and the OpenStreetMap road network; later rebuilt as a tested package with an interactive web version

Skills

Mechanical Design: SolidWorks, Onshape, GD&T, ISO 2768 Drawings, Fits & Clearances, DFM, Shaft/Coupling & Bearing Design, Gear Trains, 3D Printing & Slicing
Actuation & Mechanisms: MR-Fluid Clutch Actuators, Magnetic-Circuit Reasoning, Linkage Design, Servo Control
Electrical & Embedded: Arduino, Motor Drivers, PWM Control, Schematics & Connector Pinouts, Soldering
Robotics & Simulation: ROS 2, Gazebo, MATLAB, Simulink
Programming & Tools: Python, C/C++, Git/GitHub, Java

Awards & Certifications

Certified SolidWorks Associate (CSWA)
Dean's Honour List, Western University
3rd Place, Western Engineering Competition, Junior Division (Nov. 2025)
Top 3 of 72 Teams, Ivey Case Competition (Mar. 2025)
Other: Member, Muslim Students' Association; Full G Driver's Licence